

FOUNDATIONAL ASPECTS OF THEORIES OF MEASUREMENT

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1. Definition of Measurement. It is a scientific platitude that there can be neither precise control nor prediction of phenomena without measurement. The point of a theory of measurement is to make explicit the structure that a set of empirical data must satisfy in order to apply numerical computations to them. From an abstract standpoint a set of empirical data consists of a collection of relations between a specified set of objects. The problem of measurement is to assign numbers to the objects in such a way that the relations receive an exact and reasonable numerical interpretation.

In constructing a particular theory of measurement it is not appropriate to consider only a single set of data, for the theory should be applicable to many situations. Furthermore the theory is not concerned with all possible sets of data on relevant sets of objects, but only with those which have a structure fixed in advance. Finally coherence is given to the theory by specifying a uniform numerical interpretation of this structure.

Before turning to problems connected with construction of theories of measurement, we want to give a precise set-theoretical meaning to the notions involved. To begin with, we treat sets of empirical data as being (finitary) relational systems, that is to say, finite sequences of the form

$\mathcal{R} = \langle A, R_1, \dots, R_n \rangle$, where A is a non-empty set of elements called the domain of the relational system $\mathcal{R}$, and $R_1, \dots, R_n$ are finitary relations on A . The relational system $\mathcal{R}$ is called finite if the set A is finite;

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otherwise, infinite. It should be obvious from this definition that we are mainly considering qualitative empirical data. Intuitively we may think of each particular relation R_i (an m_i -ary relation, say) as representing a complete set of "yes" or "no" answers to a question asked of every m_i -termed sequence of objects in A . The point of this paper is not to consider that aspect of measurement connected with the actual collection of data, but rather the analysis of relational systems and their numerical interpretations.

If $s = \langle m_1, \dots, m_n \rangle$ is an n -termed sequence of positive integers then a relational system $\mathcal{R} = \langle A, R_1, \dots, R_n \rangle$ is of type s if for each $i=1, \dots, n$ the relation R_i is an m_i -ary relation. Two relational systems are similar if there is a sequence s of positive integers such that they are both of type s . Notice that type of a relational system is uniquely determined only if all the relations are non-empty; the avoiding of this ambiguity is not worthwhile. Suppose that two relational systems $\mathcal{R} = \langle A, R_1, \dots, R_n \rangle$ and $\mathcal{L} = \langle B, S_1, \dots, S_n \rangle$ are of type $s = \langle m_1, \dots, m_n \rangle$. Then $\mathcal{L}$ is a homomorphic image of $\mathcal{R}$ if there is a function f from A onto B such that for each $i=1, \dots, n$ and for each sequence $\langle a_1, \dots, a_{m_i} \rangle$ of elements of A

$$R_i(a_1, \dots, a_{m_i}) \text{ if and only if } S_i(f(a_1), \dots, f(a_{m_i})).$$

If the function f is one-one, then $\mathcal{L}$ is an isomorphic image of $\mathcal{R}$ or simply $\mathcal{R}$ and $\mathcal{L}$ are isomorphic. $\mathcal{R}$ is a subsystem of $\mathcal{L}$ if $A \subseteq B$ and for each $i=1, \dots, n$, the relation R_i is the restriction of the relation S_i to A . $\mathcal{R}$ is imbeddable in $\mathcal{L}$ if some subsystem of $\mathcal{L}$ is a homomorphic

image of $\mathcal{R}$.^{2/} A numerical relational system is simply a relational system whose domain of elements is the set $\mathcal{R}$ of all real numbers. A numerical assignment for a relational system $\mathcal{R}$ with respect to a numerical relational system $\mathcal{N}$ is a function which imbeds $\mathcal{R}$ in $\mathcal{N}$. A numerical assignment is not required to be one-one.

Within the framework of the preceeding formal definitions it is now possible to give an exact characterization of a theory of measurement. First of all the general outlines of a theory are determined by fixing a finite sequence s of positive integers and only considering relational systems of type s . Next a numerical relational system $\mathcal{N}$ of type s is selected which corresponds to the intended numerical interpretation of the theory, and only relational systems imbeddable in $\mathcal{N}$ are permitted. Moreover the theory need not concern all relational systems of type s imbeddable in $\mathcal{N}$ but only a distinguished subclass. Since it is reasonable that no special set of objects be preferred, we require that the distinguished subclass be closed under isomorphism. Indeed, those who desire to characterize theories of measurement as definite entities may actually identify each theory with its distinguished class of relational systems; to be more specific, it might be said that a theory of measurement is a class K of relational systems closed under isomorphism for which there exists a finite sequence s of positive integers and a numerical relational system $\mathcal{N}$ of type s such that all

^{2/} Although in most mathematical contexts imbeddability is defined in terms of isomorphism rather than homomorphism, for theories of measurement this is too restrictive.

relational systems in K are of type s and imbeddable in $\mathcal{N}$.^{3/} Though this definition has the advantage of formal correctness and even some intuitive justification, we do not maintain that this is the only possible definition of a theory of measurement. But when no definition is given at all it is hardly possible to ask, let alone answer, many interesting and natural questions. For this reason, if for no other, we are willing to commit ourselves to the proposed definition.

Some readers may object that the definition of theories of measurement should be linguistic rather than set-theoretical in character, since a theory is ordinarily thought of as a linguistic entity. To be sure, many theories of measurement have a natural formalization in first-order predicate logic with identity. Notice, however, that first-order axioms by themselves are not adequate, for if they admit one infinite relational system as a model then they have models of every infinite cardinality, and it is difficult to see how any natural connection can be established between numerical models and models of arbitrary cardinality. Even neglecting this criticism first-order axioms are not adequate to express properties involving arbitrary natural numbers; for example, the fact that a relational system is finite or that as an ordering it has Archimedean properties. Any linguistic definition of theories which will permit expression of these more general properties would require extensive machinery and be immediately involved

^{3/} In some contexts we shall say that the class K is a theory of measurement of type s relative to $\mathcal{N}$. Notice that a consequence of this definition is that if K is a theory of measurement, then so, is every subclass of K closed under isomorphism. Moreover, the class of all systems imbeddable in members of K is also a theory of measurement.

in some of the deepest problems of modern metamathematics. On the other hand, we do not wish to give the impression that we reject any linguistic questions. In fact, we use our set-theoretical definition as a point of departure for asking just such questions.

On the basis of the definition of theories of measurement adopted, two questions naturally arise, to each of which we devote a section. In the first place, is a given class of relational systems a theory of measurement? And in the second place, given a theory of measurement in what sense can it be axiomatized?

We would like to record here our indebtedness to Professor Alfred Tarski whose clear and precise formulation of the mathematical theory of models^{4/} has greatly influenced our presentation. Although our theories of measurement do not constitute special cases of the arithmetic classes of Tarski, the notions are closely related, and we have made use of results and methods from the theory of models.

2. Existence of Measurement. A simple counterexample shows that not every class of relational systems of a given type closed under isomorphism is a theory of measurement. Let $\mathcal{O}$ be the class of all relational systems of type $\langle 2 \rangle$ that are simple orderings. Let $\langle A, R \rangle$ be a system in $\mathcal{O}$ where R well-orders A and A has a power not equal to or less than that of the continuum. Such a relational system can be proved to exist even without the help of the axiom of choice, but of course with aid of this axiom the existence is obvious. By way of contradiction suppose that $\mathcal{O}$ is a theory of measurement

^{4/} See Tarski [6].

relative to a numerical relational system $\langle Re, S \rangle$. From the definition it follows that $\langle A, R \rangle$ is imbeddable in $\langle Re, S \rangle$ and that there is a numerical assignment f mapping A onto a subset of Re such that

$$xRy \text{ if and only if } f(x) S f(y)$$

for all elements $x, y \in A$. Let a, b be elements of A such that $f(a) = f(b)$. From the hypothesis that R is a simple ordering, we can assume without loss of generality that aRb . Hence, we have $f(a) S f(b)$, and then $f(b) S f(a)$, and finally bRa . R is antisymmetric, and so $a=b$. This argument shows that the function f is one-one. Whence, A has the same power as a subset of Re , which is impossible. This proof shows that every theory of measurement included in the class $\mathcal{O}$ contains only relational systems of power at most that of the continuum. It is an unsolved problem of set-theory closely connected with the continuum hypothesis whether the class $\mathcal{O}$ restricted to systems of power at most that of the continuum is actually a theory of measurement.^{5/} At least it can be very easily shown that $\mathcal{O}$ so restricted is not a theory of measurement relative to the system $\langle Re, \leq \rangle$, where the relation $\leq$ is the usual ordering of the real numbers.^{6/} Indeed, the exact condition that a relational system in $\mathcal{O}$ must satisfy to be imbeddable in $\langle Re, \leq \rangle$, is not really

^{5/} In this connection see Sierpinski [4] Section 7, page 141 ff., in particular Proposition C₇₅, where of course different terminology is used.

^{6/} It is sufficient here to consider a relational system isomorphic to the ordering of the ordinals of the second number class or to the lexicographical ordering of all pairs of real numbers.

elementary and the proof of the necessity involves the axiom of choice.^{7/}

Let $\mathcal{O}'$ be $\mathcal{O}$ restricted to countable^{8/} relational systems. It was proved by Cantor that $\mathcal{O}'$ is a theory of measurement relative to $\langle \text{Re}, \leq \rangle$, to formulate somewhat irreverently his classical result in the terminology of this paper. This restriction to countable relational systems is always sufficient. For it can be shown that the class of all countable relational systems of a given type is a theory of measurement; however, the numerical relational system required is so bizarre as to be of no practical value.

One of the aims of measurement is to provide a means of convenient computation. But among the morass of all possible numerical relational systems only a very few are of any computational value, indeed only those definable in terms of the ordinary arithmetical notions. From an empirical standpoint most sets of qualitative data can find numerical interpretation by relations defined in terms of addition and ordering alone. By way of example we may cite the measurement of masses, distances, sensation intensities, and subjective probabilities. Frequently the consideration of weighted averages requires also the use of the multiplication of numbers. However, in the examples given in this paper we shall restrict ourselves to the notions of addition and ordering.

No natural scientific situation would seem strictly to require the consideration of sets of infinite data. This state of affairs suggests that theories of measurement containing only finite relational systems would

^{7/} A simple ordering is imbeddable in $\langle \text{Re}, \leq \rangle$ if and only if it contains a countable dense subset. For the exact formulation and a sketch of a proof see Birkhoff [1], pp. 31-32, Theorem 2.

^{8/} The word 'countable' means at most denumerable and it refers to the cardinality of the domains of the relational systems.

suffice for empirical purposes. The problem is delicate, however, for the measurement of a meteorological quantity such as temperature by an automatic recording device is usually treated as continuous both in its own scale and in time. Yet the important problem of measurement does not really lie in the correct use of such recording devices but rather in their initial calibration, a process proceeding from a finite number of qualitative decisions. Because of the awkwardness of the uniform application of finite relational systems, we shall not generally make this restriction.

Further remarks about establishing the existence of measurement are best motivated by reference to a concrete example. In a recent paper^{9/} R. D. Luce has introduced a generalization of simple orderings which he calls semiorders. A semiorder is a relational system $\langle A, P \rangle$ of type $\langle 2 \rangle$ which satisfies the following axioms for all $x, y, z, w \in A$:

S1. Not xPx .

S2. If xPy and zPw , then either xPw or zPy .

S3. If xPy and zPx , then either wPy or zPw .^{10/}

Such relations are most likely to occur in situations where objects are to be arranged in order and where it is difficult to say exactly when two objects are indifferent. For example, to say that xPy might be interpreted as meaning that the pitch of the sound x is definitely higher than the pitch

^{9/} See Luce [3].

^{10/} See Luce [3] Section 2, p. 181. The axioms given here are actually a simplification of those given by Luce.

of y , or that the hue of color x is definitely brighter than the hue of color y , or that the weight of the object x is noticeably greater than that of y , etc. Indifference between two objects x and y (in symbols: xIy) is defined as not xPy , and not yPx . The point of Luce's axioms is that the relation I of indifference is not always transitive, a fact easily appreciated for each of the intuitive interpretations given above.

In his paper Luce gives a certain numerical interpretation for certain kinds of semiorders, but he does not show that any particular class of semiorders is a theory of measurement in the sense used here, because his interpretations are not relative to a fixed numerical relation. In the denumerable case the situation is very simple indeed. Let $\gg$ be the relation between real numbers defined by the condition

$$x \gg y \text{ if and only if } x > y + 1.$$

Obviously, if x and y are real numbers such that $x \gg y$, then x is definitely greater than y or better x is noticeably greater than y . The following result is not difficult to establish.

The class of countable semiorders is a theory of measurement relative to the numerical relational system $\langle \text{Re}, \gg \rangle$ (which is itself a semiorder).

It is possible to characterize all semiorders imbeddable in $\langle \text{Re}, \gg \rangle$, similar to the result for simple orderings, but this shall not concern us here.

For the remainder of this discussion let C be the class of all countable semiorders. The class C affords an example of a theory of measurement (relative to $\langle \text{Re}, \gg \rangle$) for which not all numerical assignments are one-one functions.^{11/} However, this theory has the peculiarity that those systems

^{11/} For example, consider any semiorders in which any two elements are indifferent.

having only one-one assignments can be distinguished by a simple additional axiom. Let $\langle A, P \rangle$ be any semiorder. Define a relation E between elements of A by the condition

xEy if and only if for all $z \in A$ the formulas xIz and yIz are equivalent.

The relation E is easily proved to be an equivalence relation, and, in fact, it is a congruence relation in the sense that for all $x, y, z \in A$

if xPz and xEy then yPz , and

if zPx and xEy , then zPy .

Furthermore, we can show that if f is any numerical assignment imbedding $\langle A, P \rangle$ in $\langle \mathbb{R}, > \rangle$ then for all $x, y \in A$, if $f(x) = f(y)$, then xEy . Consider then the axiom:

S4. If xEy , then $x = y$.

Let C^* be the class of all systems in C which satisfy S4. Clearly from our remarks above it follows that any system in C^* has only one-one numerical assignments. In the other direction, assume that $\langle A, P \rangle$ is a system C possessing only one-one assignments. Since the relation E is a congruence relation for the system $\langle A, P \rangle$, we can reduce by E and form a system of cosets $\langle A^*, P^* \rangle$, where A^* is the family of equivalence classes under E , and P^* is defined in the natural way. Obviously $\langle A^*, P^* \rangle$ is in C^* and $\langle A, P \rangle$ is imbeddable in $\langle A^*, P^* \rangle$ (in fact, $\langle A^*, P^* \rangle$ is a homomorphic image of $\langle A, P \rangle$). Let f^* be any assignment for $\langle A^*, P^* \rangle$. In a straightforward way we can define an assignment f for $\langle A, P \rangle$ such

that $f(x) = f^*([x])$, where $x \in A$ and $[x]$ is the E-equivalence class of x . Now by assumption f is one-one. Hence if xEy , then $[x] = [y]$, and so $f^*([x]) = f^*([y])$, which implies $f(x) = f(y)$ and $x = y$. Thus $\langle A, P \rangle$ itself satisfies S4. We have shown that C^* is exactly the class of all systems in C possessing only one-one assignments.

To be truthful, the above discussion has somewhat inverted the natural sequence of steps. In order to establish the existence of measurement for C (that is, to show that C is a theory of measurement relative to $\langle Re, >> \rangle$), it is far better to consider the class C^* first and prove that it is a theory of measurement relative to $\langle Re, >> \rangle$. Finally it need only be remarked that every system in C can be imbedded in a system of C^* by the method of cosets. As a matter of fact it can also be proved that every system in C is a subsystem of some system in C^* , which is another method of proving that C is a theory of measurement.

Let us now summarize the steps in establishing the existence of measurement using as models the simple orderings and the semiorders. First, after one is given a class, K say, of relational systems, the numerical relational system should be decided upon. The numerical relational system should be naturally suggested by the structure of the systems in K , and as was remarked, it is most practical to consider numerical systems where all the relations can be simply defined in terms of addition and ordering of real numbers. Second, if the proof that K is a theory of measurement is not at once obvious, the cardinality of systems in K should be taken into consideration. The restriction to countable systems would always seem empirically justified, and adequate results are often possible with a restriction to finite systems. Third, the

proof of the existence of measurement can often be simplified by the reduction of each relational system in K by the method of cosets. One usually looks for a uniform and natural method of introducing into each relational system in K an equivalence relation which preserves all the relations between the elements. Then, instead of trying to find numerical assignments for each member of K , one concentrates only on the reduced systems. This plan was very helpful in the case of semiorders. Instead of cosets, it is sometimes feasible to consider imbedding by subsystems. That is to say, one considers some convenient subclass $K' \subseteq K$ such that every element of K is a subsystem of some system in K' . If K' is a theory of measurement then so is K . In the case of semiorders we could have used either plan: cosets or subsystems.

After the existence of measurement has been established, there are two questions which are often of interest: for a given relational system what is the class of all its numerical assignments? and how can the class of all systems isomorphic to the given system be described? These two questions are not unrelated, though in some cases one can be answered without deciding the other. We shall present two examples.

Consider relational systems $\langle A, D \rangle$ of type $\langle 4 \rangle$. For such systems we introduce the following definitions:

xRy if and only if $xyDyy$

xyM^1_{zw} if and only if $xyDzw, zwDxy, yRz$ and zRy

xyM^{n+1}_{zw} if and only if there exist

$u, v \in A$ such that

xyM^n_{uv} and uvM^1_{zw} .

Let H be the class of all relational systems $\langle A, D \rangle$ of type $\langle 4 \rangle$ which satisfy the following axioms for every $x, y, z, u, v, w \in D$:

- A1. If $xyDzw$ and $zwDuv$ then $xyDuv$.
- A2. $xyDzw$ or $zwDxy$.
- A3. If $xyDzw$ then $xzDyw$.
- A4. If $xyDzw$ then $wzDyx$.
- A5. If xRy and $yzDuv$ then $xzDuv$.
- A6. There is a $z \in D$ such that $xzDzy$ and $zyDxy$.
- A7. If not $xyDzw$ and not xRy then there is a $u \in D$ such that $zWDxu$, not xRu and not uRy .
- A8. If $xyDzw$ and not xRy then there are $u, v \in D$ and an n such that $zuM^n vw$ and $zuDxy$.

These axioms imply that for a system $\langle A, D \rangle$ in H , the relation R is a weak ordering of A , and the intuitive interpretation of $xyDzw$ in case yRx and wRz is that the interval between x and y is not greater than the interval between z and w . Making heavy use of the last three existence axioms (note that A8 gives an Archimedean property to the ordering), it can be shown that H is a theory of measurement relative to the numerical relational system $\langle Re, \Delta \rangle$ where Δ is the quaternary relation defined by the condition.

$$xy\Delta zw \text{ if and only if } x-y \leq z-w$$

for all $x, y, z, w \in Re$. In addition, it can be shown that if $\langle A, D \rangle$ is in

H and f and g are two numerical assignments of $\langle A, D \rangle$ relative to $\langle Re, \Delta \rangle$, then f and g are related by a positive linear transformation;^{12/} that is, there exist $\alpha, \beta \in Re$ with $\alpha > 0$ such that for all $x \in Re$, $f(x) = \alpha g(x) + \beta$. This gives in a certain sense the answer to the first question above: if we know one numerical assignment of $\langle A, D \rangle$, we know them all. Except for very special systems in H , nothing more specific can really be expected.

Notice that all relational systems in H are necessarily infinite. In the next section we shall consider in detail the theory of measurement F consisting of all finite relational systems imbeddable in $\langle Re, \Delta \rangle$. Here the situation is quite hopeless. There simply is no apparent general statement that can be made about the relation between assignments. In as much as any function ϕ which imbeds $\langle Re, \Delta \rangle$ in itself is necessarily a linear transformation and conversely, it follows that if $\langle A, D \rangle$ is a system in F and f is an assignment for $\langle A, D \rangle$, then f composed with a linear transformation is also an assignment. The main difficulty with F is that two assignments for the same system in F need not be related by a linear transformation.

The situation with regard to the class C^* of semiorders introduced above is much the same as that for F , in as much as there is no apparent relation between assignments. However, though it seems difficult to describe all the different finite systems in F , we can answer this question for the finite systems in C^* . The description is facilitated by the following definition.

^{12/} The proofs of both these facts about H are very similar to the corresponding proofs in Suppes and Winet [5].

Let $s = \langle s_1, \dots, s_n \rangle$ and $t = \langle t_1, \dots, t_n \rangle$ be two finite sequences of positive integers of the same length n . We write $s \leq t$ to mean:

- (i) $s_{i+1} > s_i$ for $i < n$,
- (ii) $t_{i+1} > t_i$ for $i < n$,
- (iii) $s_i > t_i$ for $i \leq n$.

Let m be any integer not less than s_n . We define $\mathcal{O}_m(s, t)$, where $s \leq t$, to be the relational system $\langle A, P \rangle$, where $A = \{1, \dots, m\}$ and the binary relation P is defined by the condition

$kP\ell$ if and only if there is an integer $i \leq n$ such that

$$k \geq s_i > t_i \geq \ell$$

for all $k, \ell \in A$.

It is possible without much trouble to establish the following facts:

- (1) $\mathcal{O}_m(s, t)$ is a semiorder;
- (2) If $\mathcal{O}_m(s, t)$ satisfies axiom S4, and if $\mathcal{O}_m(s, t)$ is isomorphic to $\mathcal{O}_{m'}(s', t')$, then $m = m'$, $s = s'$, and $t = t'$;
- (3) Any finite semiorder is isomorphic to a system $\mathcal{O}_m(s, t)$ for suitable m, s , and t .

Thus what has been accomplished is the exhibition of exactly one representative of each isomorphism type of finite semiorders which satisfy axiom S4. Actually, with very little more effort we could do the same for arbitrary finite semiorders. Since a system $\mathcal{O}_m(s, t)$ seems to be a definite object that we can constructively define, we feel that this construction yields an adequate

description of finite semiorders and answers the second of our questions.

3. Axiomatizability. Given a theory of measurement it is natural to ask various questions about its axiomatizability, for the axiomatic analysis of any mathematical theory usually throws considerable light on the structure of the theory. In particular given an extrinsic characterization of a theory of measurement via a particular numerical relational system, it is quite desirable to have an intrinsic axiomatic characterization of the theory to be able better to recognize when a relational system actually belongs to the theory. In view of the paucity of metamathematical results concerning the axiomatics of higher-order theories, we shall restrict ourselves to the problem of axiomatizing theories of measurement in first-order logic.

It is a well-known result that if a set of first-order axioms has one infinite model, then it has models of unbounded cardinalities. Since for the most part we are interested in one-one assignments with values in the set of real numbers, unbounded cardinalities are hardly an asset. That is to say: the class of all relational systems that are models of a given set of first-order axioms is usually not a theory of measurement. To remove such difficulties without having to understand them, we simply restrict the cardinalities under consideration. Even a restriction to finite cardinalities is not too strong and leads to some rather difficult questions. Thus for the remainder of this section we shall consider only finitary theories of measurement, i.e., theories containing only finite relational systems. Such a theory is called axiomatizable if there exists a set of sentences of first-order logic (the axioms of the theory) such that a finite relational system is in the theory if and only if the system satisfies all the sentences in the set.

A theory is finitely axiomatizable if it has a finite set of axioms. A theory is universally axiomatizable if it has a set of axioms each of which is a universal sentence (i.e., a sentence in prenex normal form with only universal quantifiers).

It should be observed first that any finitary theory of measurement is axiomatizable. This is no deeper than saying that in first-order logic we can write down a sentence completely describing the isomorphism type of each finite relational system in the given theory, and clearly these sentences can serve as the required set of axioms. It is of course quite obvious that we cannot give in each instance an effective method for writing down the axioms, since there are clearly a continuum number of distinct finitary theories of measurement. Notice also that if the theory is closed under subsystems then the axioms may be taken as universal sentences, and conversely. In case one considers theories consisting of all finite relational systems imbeddable in a given numerical relational system, then the problem of a recursive or effective axiomatization is simply the problem of whether the class of universal sentences true in the given numerical relational system is recursively enumerable or not. It is not difficult to establish that this last problem is equivalent to the problem of giving a recursive enumeration of all the relation types of finite relational systems not imbeddable in the given numerical relational system. For numerical relational systems whose relations are definable in first-order logic in terms of $+$ and $\leq$, these problems do not arise since the first-order theory of $+$ and $\leq$ is decidable, and it is to these relational systems that we shall primarily restrict our further attention.

The most interesting axiomatizability question for finitary theories of measurement would seem to be: when are they finitely axiomatizable? In empirical applications a finite axiomatization provides the basis for an exact classification of sources of error. And if the axioms are universal sentences, the computational decision as to whether a given relational system is a model of the axioms (and thus a member of the theory of measurement defined by the axioms) reduces to considering subsystems of the given relational system which have a cardinality equal to the number of distinct variables required to write the axioms as a single universal sentence. Vaught [7] has provided a useful criterion for certain classes of relational systems to be axiomatizable by means of a universal sentence. His result yields immediately the following criterion for finitary theories of measurement.

A finitary theory of measurement K is axiomatizable by a universal sentence if, and only if, K is closed under subsystems and there is an integer n such that if any finite relational system $\mathcal{R}$ has the property that every subsystem of $\mathcal{R}$ with no more than n elements is in K then $\mathcal{R}$ is in K.

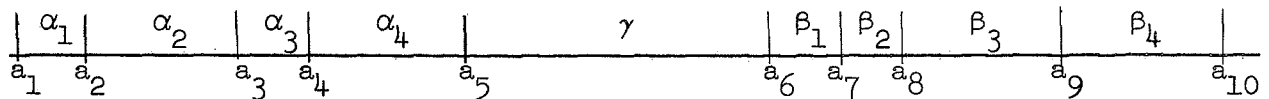
The classes of finite simple orderings and finite semiorders are two examples of finitary theories of measurement axiomatizable by a universal sentence. On the other hand there are interesting examples of finitary theories of measurement closed under subsystems which are not axiomatizable by a universal sentence. We now turn to one such example.

Let F be the class of all finitary relational systems of type $\langle 4 \rangle$ imbeddable in the numerical relational system $\langle Re, \Delta \rangle$. A wide variety of sets of empirical data are in F . In fact, all sets of psychological data

based upon judgments of differences of sensation intensities or of differences in utility qualify as candidates for membership in F . For example, in an experiment concerned with the subjective measurement of loudness of n sounds, the appropriate empirical data would be obtained by asking subjects to compare each of the n sounds with every other and then to compare the difference of loudness in every pair of sounds with every other. More elaborate interpretations are required to obtain appropriate data on utility differences for individuals or social groups (cf. Davidson, Suppes and Siegel [2], Suppes and Winet [5]). It may be of some interest to mention one probabilistic interpretation closely related to the classical scaling method of paired comparisons. Subjects are asked to choose only between objects, but they are asked to make this choice a number of times. There are many situations in which they vacillate in their choice, and the probability p_{xy} that x will be chosen over y may be estimated from the relative frequency with which x is so chosen. From inequalities of the form $p_{xy} \leq p_{zw}$ we may obtain a set of empirical data, that is, a finite relational system of type $\langle 4 \rangle$, which is a candidate for membership in F . The intended interpretation is that if $p_{xy} \geq \frac{1}{2}$ and $p_{zw} \geq \frac{1}{2}$, then $p_{xy} \leq p_{zw}$ if and only if the difference in sensation intensity or difference in utility between x and y is equal to or less than that between z and w , the idea being, of course, that if x and y are closer together than z and w in the subjective scale, then the relative frequency of choice of x over y is closer to one-half than that of z over w .

Before formally proving that the theory of measurement F is not axiomatizable by a universal sentence, we intuitively indicate for a

relational system of ten elements the kind of difficulty which arises in any attempt to axiomatize F . Let the ten elements be $a_1, \dots, a_{10}$ ordered as shown on the following diagram with atomic intervals given the designations indicated.



Let α be the interval (a_1, a_5) , let β be the interval (a_6, a_{10}) , and let γ be larger than α or β . We suppose further that

α_1 is equal in size to β_2

α_2 is equal in size to β_4

α_3 is equal in size to β_1

α_4 is equal in size to β_3

but

α is less than β .^{13/}

The size relationships among the remaining intervals may be so chosen that any subsystem of nine elements is imbeddable in $\langle \text{Re}, \Delta \rangle$, whereas the full system of ten elements is clearly not.

Generalizing this example and using the criterion derived from Vaught's

^{13/} Essentially this example was first given in another context by Herman Rubin to show that a particular set of axioms was defective.

theorem we now prove:

Theorem: The theory of measurement F is not axiomatizable by a universal sentence.

Proof: In order to apply the criterion of axiomatizability by a universal sentence, we need to show that for every n there is a finite relational system $\mathcal{M}$ of type $\langle 4 \rangle$ such that every subsystem of $\mathcal{M}$ with n elements in its domain is in F but $\mathcal{M}$ is not.

To this end, for every even integer $n \geq 10$ we construct a finite relational system $\mathcal{M}$ of type $\langle 4 \rangle$ such that every subsystem of $n-1$ elements is in F . (A fortiori every subsystem of $n-k$ elements for $k < n$ is in F .) To make the construction both definite and compact, we take numbers as elements of the domain and disrupt exactly one numerical relationship. Let now n be an even integer equal to or greater than 10. The selection of numbers $a_1, \dots, a_n$ may be most easily described by specifying the numerical size of the atomic intervals. We define:

$$\alpha_i = a_{i+1} - a_i \quad \text{for } i=1, \dots, \frac{n}{2} - 1$$

$$\beta_i = a_{\frac{n}{2}+i+1} - a_{\frac{n}{2}+i} \quad \text{for } i=1, \dots, \frac{n}{2} - 1.$$

We then set:

$$\alpha_i = 2^i \quad \text{for } i=1, \dots, \frac{n}{2} - 1$$

$$a_{\frac{n}{2}+1} = 2^n.$$

In fixing the size of β_i , we have two cases to consider depending on the

parity of $\frac{n}{2}$.

Case 1. $\frac{n}{2}$ is even. Then $\frac{n}{2} - 1$ is odd, and we set: if i is even

$$\beta_i = \alpha_{\frac{i}{2}} \quad \text{for } i=2,4,\dots,\frac{n}{2} - 2$$

and if i is odd

$$\beta_i = \alpha_{\frac{n}{4} + \frac{i-1}{2}} \quad \text{for } i=1,3,\dots,\frac{n}{2} - 1.$$

Case 2. $\frac{n}{2}$ is odd. Then $\frac{n}{2} - 1$ is even, and we set: if i is even

$$\beta_i = \alpha_{\frac{i}{2}} \quad \text{for } i=2,4,\dots,\frac{n}{2} - 1$$

and if i is odd

$$\beta_i = \alpha_{\frac{n-2}{4} + \frac{i+1}{2}} \quad \text{for } i=1,3,\dots,\frac{n}{2} - 2.$$

Thus if $n=12$, we have:

$$\alpha_1 = \beta_2$$

$$\alpha_2 = \beta_4$$

$$\alpha_3 = \beta_1$$

$$\alpha_4 = \beta_3$$

$$\alpha_5 = \beta_5.$$

With the set $A = \{a_1, \dots, a_n\}$ defined, we now define the relation D

as the expected numerical relation except for permutations of $a_1, a_{\frac{n}{2}}, a_{\frac{n}{2}+1}$ and a_n . If $x, y, z, w \in A$ and $\langle x, y, z, w \rangle$ is not some permutation of $\langle a_1, a_{\frac{n}{2}}, a_{\frac{n}{2}+1}, a_n \rangle$ then $\langle x, y, z, w \rangle \in D$ if and only if

$$(1) \quad x - y \leq z - w.$$

Moreover, let $a = a_1, b = a_{\frac{n}{2}}, c = a_{\frac{n}{2}+1}, d = a_n$, then we put the following nine permutations of $\langle a, b, c, d \rangle$ in D :

$$(2) \quad \begin{array}{ll} \langle b, a, d, c \rangle & \langle a, d, c, b \rangle \\ \langle b, d, a, c \rangle & \langle c, b, d, a \rangle \\ \langle b, d, c, a \rangle & \langle c, d, a, b \rangle \\ \langle a, b, d, c \rangle & \langle c, d, b, a \rangle \\ \langle a, c, d, b \rangle & \end{array}$$

(These nine permutations just correspond to the strict inequalities following from $b - a < d - c$. All nine are needed to make the subsystems of $\langle A, D \rangle$ have the appropriate properties.)

From the choice of the numbers in A and the definition of D it is obvious that $\langle A, D \rangle$ is not imbeddable in $\langle \mathbb{R}, \Delta \rangle$, that is, that $\langle A, D \rangle$ is not in F , for the atomic intervals between a_1 and $a_{\frac{n}{2}}$ must add up to a length equal to the sum of the atomic intervals between $a_{\frac{n}{2}+1}$ and a_n , but by hypothesis the interval $(a_1, a_{\frac{n}{2}})$ is less than the interval $(a_{\frac{n}{2}+1}, a_n)$. It remains to show that every subsystem of $n-1$ elements is in F . Two cases

naturally arise.

Case 1. The element omitted in the subsystem is $a_1, a_{\frac{n}{2}}, a_{\frac{n}{2}+1}$ or a_n .

Then the nine permutations of (2) are not in D restricted the subsystem, and the subsystem is not merely imbeddable in $\langle Re, \Delta \rangle$ but by virtue of (1) is a subsystem of it.

Case 2. The element omitted is neither $a_1, a_{\frac{n}{2}}, a_{\frac{n}{2}+1}$ nor a_n . Let a_i be the element not in the subsystem. There are two cases to consider.

Case 2a. $a_i < a_{\frac{n}{2}}$. For this situation we may use for our numerical assignment the function f defined as follows:

$$f(a_{i-j}) = a_{i-j} + 1 \quad \text{for } j=1, \dots, i-1$$

$$f(a_{i+j}) = a_{i+j} \quad \text{for } j=1, \dots, n-i.$$

It is straightforward but tedious to verify that f is a numerical assignment, that is, that it preserves the relation D as defined by (1) and (2). Only two observations are crucial to this verification. First, regarding atomic intervals (in the full system), if

$$a_{i-j+1} - a_{i-j} = a_{k+1} - a_k \quad \text{for } k > i,$$

then

$$\begin{aligned} f(a_{i-j+1}) - f(a_{i-j}) &= (a_{i-j+1} - 1) - (a_{i-j} - 1) \\ &= a_{k+1} - a_k \\ &= f(a_{k+1}) - f(a_k). \end{aligned}$$

Second, the numbers in A were so chosen that if $x, y, z, w \in A$, if (z, w) is not an atomic interval, if $(x, y) \neq (z, w)$ and if

$$x - y \leq z - w,$$

then

$$(3) \quad x - y + 2 \leq z - w;$$

whence it is clear from the definition of f that

$$f(x) - f(y) \leq f(z) - f(w).$$

(Note that (3) implies the weaker result that no two distinct non-atomic intervals have the same size.)

Case 2b. $a_i > a_{\frac{n}{2}} + 1$. Here we may use a numerical assignment f

defined, as would be expected from the previous case, by the equations:

$$f(a_{i-j}) = a_{i-j} \quad \text{for } j=1, \dots, i-1$$

$$f(a_{i+j}) = a_{i+j} + 1 \quad \text{for } j=1, \dots, n-i.$$

And this completes the proof of the theorem.

It would be pleasant to report that we could prove a stronger result about the theory of measurement F , namely, that it is not finitely axiomatizable. Unfortunately, there seems to be a paucity of tools available for studying such questions for classes of relational systems. However, we would like to state a conjecture which if true would provide one useful tool for studying the finite axiomatizability of finitary theories of measurement like F

which are closed under submodels. We say that two sentences are finitely equivalent if and only if they are satisfied by the same finite relational systems, and we conjecture: if S is a sentence such that if it is satisfied by a finite model it is satisfied by every submodel of the finite model, then there is a universal sentence finitely equivalent to S . If this conjecture is true it follows that any finitary theory of measurement closed under submodels is finitely axiomatizable if and only if it is axiomatizable by a universal sentence.

The proof (or disproof) of this conjecture appears difficult. It easily follows from Tarski's results [6] on universal (arithmetical) classes in the wider sense that if the finitistic restrictions are removed throughout in the conjecture, the thus modified conjecture is true. For S , being closed under submodels, defines a universal class in the wider sense of relational systems, which class is axiomatizable by a denumerable set of universal sentences. Since S is logically equivalent to this set of universal sentences, it is a logical consequence of some finite subset of them, but because it implies the full set, it also implies the finite subset and is thus equivalent to it.

Our conjecture is one concerning the general theory of models and its pertinence is not restricted to theories of measurement. In conclusion we should like to mention two unsolved problems typical of those which arise in the special area of measurement. (i) Let R be any binary numerical relation definable in an elementary manner in terms of plus and less than. Is every finitary theory of measurement with respect to R finitely axiomatizable? (If our conjecture about finite models is true then the theory of measurement F is not finitely axiomatizable and shows that the answer to this

problem is negative for quaternary relations definable in terms of plus and less than.) (ii) Is every finitary theory of measurement axiomatizable by a universal sentence a theory of measurement with respect to a numerical relational system all of whose relations are definable in terms of addition and ordering of real numbers?

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